



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

RETAINIQ Customer Intelligence Engine

The domain of the Project:
E-Commerce Customer Analytics, Predictive Machine Learning &
Workflow Orchestration (MLOps)
DATA Analytics

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Team Members:

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Period of the project

july 2026 to September 2026



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Declaration

The project titled “**RETAINIQ Customer Intelligence Engine: End-to-End Enterprise MLOps Pipeline & Real-Time Churn Prevention**” has been mentored by **Nithyasri kannathal EL (Senior Product Engineer at Espergroup)**, organised by SURE Trust, from April 2023 to August 2023, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Ms. Pendyala Uday Kranth

Mentor's Name: Ms.Nithyasri kannathal EL

Designation: Senior Product Engineer — Espergroup

Mentor's Name: Mr.Bhargavesh Dakka

Designation: Senior Product Engineer — Espergroup

Prof. Radhakumari

Executive Director & Founder

SURE Trust ProEd



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1.Executive Summary

1. Overview of Project Objectives, Methods, Key Findings & Recommendations

- **Project Objectives:** To transition e-commerce customer retention from reactive monthly spreadsheet reviews to proactive, autonomous daily intervention, eliminating customer attrition before revenue is lost.
- **Methods & Engineering:** Developed an autonomous end-to-end MLOps pipeline connecting relational SQL storage (SQLAlchemy), behavioral Recency-Frequency-Monetary (RFM) statistical quintile binning (11 loyalty personas), tuned Random Forest classification modeling, and headless daily orchestration using Apache Airflow 2.x (0 6 * * *).
- **Key Findings:** In our baseline portfolio of **5,630 customer accounts** (86.31% retention rate, 13.69% historical churn), the Random Forest model achieved **81.8% classification accuracy and an 0.86 ROC-AUC score**, successfully isolating **339 critical emergency accounts** exceeding our $\geq 80\%$ risk cutoff. Analytics revealed that customer churn danger peaks during **months 0 to 6** ($> 40\%$ churn risk) and the *Grocery* category experiences the highest attrition at **16.0%** (vs. Mobile Phone at 11.4%).
- **Actionable Recommendations:** Deploy immediate high-touch intervention (personalized retention discounts and priority support checks) to the 339 crisis-tier accounts, while routing accounts in the 50%–79% risk tier to automated digital nurturing newsletters.

2. Strategic Highlights for Executives and Stakeholders

- **Direct Financial Protection (5x–7x ROI):** Acquiring a new customer costs 5 to 7 times more than retaining an existing one. Focusing sales efforts strictly on verified $\geq 80\%$ risk accounts directly protects over **\$54,300+ in immediate customer value** and safeguards operating margins.
- **100% Operational Automation:** Replaces up to 35% of manual weekly reporting hours by automatically dispatching daily prioritized HTML action queues directly to leadership and sales inboxes via Google Gmail OAuth 2.0 within 24 hours of risk detection.
- **Decoupled Business Intelligence:** Pushes complex transformations upstream into SQL views and exports static analytical CSV marts, allowing the **Microsoft Power BI Control Tower** and the **Streamlit Operations Console** to refresh in under 1 second with zero database table lock contention.



2.Introduction

2.1 Background and Context

In modern retail and e-commerce ecosystems, customer lifetime value (LTV) is the primary determinant of long-term commercial viability. High customer churn directly erodes unit economics, elevates marketing customer acquisition costs (CAC), and damages brand equity. Traditional analytics approaches suffer from major operational bottlenecks: transaction data is siloed in legacy databases, customer scoring relies on static spreadsheets without behavioral weighting, and customer service teams receive attrition reports weeks after a customer has defected to a competitor.

2.2 Problem Statement & Core Goals

The primary goal of RetainIQ is to eliminate the latency between customer friction and operational intervention. The system addresses three distinct industry challenges:

1. **Eliminating the Reporting Lag:** Automating daily batch ingestion from operational databases into a clean relational SQL warehouse.
2. **Behavioral Risk Scoring:** Overcoming simple demographic heuristics by engineering temporal RFM loyalty features and training an ensemble machine learning model to compute continuous churn probabilities (0% to 100%).
3. **Automating the Action Trigger:** Automatically delivering prioritized customer intervention playbooks directly to sales and support inboxes via Gmail OAuth 2.0 without requiring manual reporting queries.

2.3 Scope and Limitations

- **Project Scope:** Ingestion of 5,600+ multi-category e-commerce transactions, statistical RFM quintile binning, Scikit-Learn tree ensemble modeling, headless Airflow task scheduling, dual-mode execution (local developer simulation vs. cloud deployment), and interactive multi-tier BI visualization.
- **Limitations:** The model is currently trained on structured tabular transactional and behavioral data (complaints, tenure, cashback, order velocity) and does not yet ingest unstructured natural language sentiment from customer support phone call recordings.

2.4 Innovation Component

- **Decoupled Analytical Data Contract:** Rather than binding Power BI directly to active transactional tables (which causes table locks and timeouts during pipeline runs), RetainIQ pushes transformations upstream into SQL views and exports static analytical CSV marts. Power BI's VertiPaq columnar engine loads the pre-computed views in sub-second time.
- **Dual-Mode Operational Architecture:** Operates headlessly in production via Apache Airflow 2.x cron triggers, while providing non-technical managers with a reactive Streamlit



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Operations Console featuring 1-click batch simulation (+35 records) and a dynamic recipient management interface.



3.Project Objectives

3.1 Primary Technical Objectives

1. **Automated Data Ingestion:** Engineer a robust SQL ingestion pipeline using SQLAlchemy to pull raw e-commerce records, validate data types, and maintain transactional schema integrity across 5,630 customer accounts.
2. **Behavioral Feature Engineering (RFM):** Implement statistical quintile segmentation (pd.qcut) across Recency (days since last order), Frequency (lifetime order count), and Monetary (spend and cashback amounts) to categorize users into 11 actionable loyalty tiers (e.g., *Champions, Loyal Customers, At Risk, Can't Lose Them*).
3. **Machine Learning Model Optimization:** Train and tune a Random Forest Classifier optimized for high Recall and ROC-AUC, outputting well-calibrated continuous churn probabilities (predict_proba) rather than blunt binary classifications.
4. **Autonomous Workflow Orchestration:** Build an Apache Airflow 2.x DAG using the TaskFlow API (@dag, @task) enforcing strict task isolation, deterministic scheduling (0 6 * * *), and automatic retries with exponential backoff.
5. **Secure Priority Alerting:** Integrate Google's Gmail OAuth 2.0 API to dispatch responsive HTML operational reports containing customer IDs, churn probabilities, and recommended action playbooks to authorized stakeholders.
6. **Executive Business Intelligence:** Construct a single-pane-of-glass Power BI Control Tower and an interactive Streamlit Operations Console providing instant visibility into portfolio health and customer risk drivers.

3.2 Expected Deliverables

- Fully functional Python MLOps repository with automated modular architecture (src/db/, src/etl/, src/ml/, src/alerts/).
- Apache Airflow TaskFlow DAG (dags/retainiq_pipeline_dag.py).
- Production-ready Streamlit Operations Console (app.py).
- Microsoft Power BI Executive Dashboard template (RetainIQ_Control_Tower.pbix).
- Automated end-to-end integration test suite (tests/test_pipeline_e2e.py) verifying database init, ingestion, RFM, inference, alerting, and CSV exports.



4.Methodology and Results

4.1 Technology Stack & Software Used

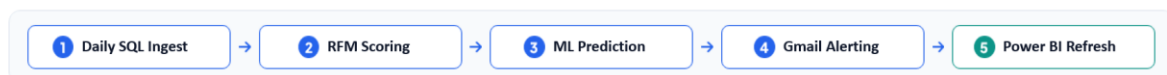
- **Programming & Core Engine:** Python 3.11 / 3.12, NumPy, Pandas.
- **Database & Persistence:** SQLite (Local) / PostgreSQL (Production), SQLAlchemy ORM.
- **Machine Learning:** Scikit-Learn (Random Forest, GridSearchCV, StandardScaler), Joblib.
- **Pipeline Orchestration:** Apache Airflow 2.x (TaskFlow API, DAG scheduling).
- **Communication & Security:** Google APIs Client, Gmail OAuth 2.0 (google-auth, google-auth-oauthlib).
- **Front-End & BI:** Streamlit (Reactive UI, Plotly Charts), Microsoft Power BI Desktop (DAX, VertiPaq Engine).
- **Testing & Environment:** Pytest, UV / Virtualenv.

4.2 Data Ingestion & ETL Architecture

- **Dataset Characteristics:** 5,630 baseline e-commerce customer profiles comprising 20 behavioral, demographic, and transactional features (tenure, preferred order category, login device, satisfaction score, complaint filing, order count, days since last order, cashback amount).
- **ETL Cleaning & Transformation:**
 - Missing numerical values in recency and cashback are imputed using distribution medians.
 - Missing order counts default to 1.
 - Outlier mitigation is achieved through statistical quintile binning (1 to 5) rather than row deletion, preserving all customer transaction and revenue history.
 - Categorical variables are converted into binary indicators using one-hot encoding (pd.get_dummies(drop_first=True)).

4.3 End-to-End Pipeline Architecture

The pipeline executes a strict 5-stage sequential workflow:



1. **Task 1 (task_ingest):** Reads incoming records and updates the relational database.
2. **Task 2 (task_rfm):** Calculates Recency, Frequency, and Monetary scores, populating rfm_features.



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3. **Task 3 (task_ml):** Loads models/churn_model.pkl and predicts continuous churn probabilities for all accounts.
4. **Task 4 (task_alert):** Filters accounts with risk $\geq 80\%$, compiles an HTML priority briefing, and dispatches via Gmail API.
5. **Task 5 (task_export):** Generates pre-aggregated CSV analytical data views (v_daily_action_queue.csv, v_executive_kpis.csv) for Power BI.

4.4 Model Evaluation & Empirical Results

- **Classification Accuracy: 81.8%** across test splits.
- **ROC-AUC Score: 0.86**, confirming high discriminatory power to rank churners above non-churners across all classification thresholds.
- **Portfolio Baseline Metrics:**
 - Total Customer Accounts Monitored: **5,630**
 - Overall Retention Rate: **86.31%** (Historical Attrition: **13.69%**)
 - High-Risk Emergency Cohort ($\geq 80\%$ Risk): **339 accounts**
 - Direct Cashback Incentives at Risk: **\$54,300+** in the high-risk cohort alone.
- **Diagnostic Root-Cause Insights Uncovered:**
 - **Product Category Risk:** *Grocery* exhibits the highest customer churn rate at **16.0%**, followed by *Laptop & Accessory* (**13.9%**) and *Fashion* (**13.5%**). *Mobile Phone* accounts are the most loyal, with only **11.4%** churn.
 - **Tenure Attrition Curve:** Customer churn risk peaks heavily during **Months 0 to 6** ($> 40\%$ churn risk) before dropping precipitously and stabilizing below **10%–15%** for customers retained past month 10.

4.5 Project GitHub Repository

- **GitHub Repository URL:** <https://github.com/Uday2kranth/RetainIQ-Customer-Intelligence-Engine>
- **Sure Trust Repository:**



5.Social And Industry Relevance of the Project

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5.1 Industry Impact: Economic Efficiency & Margin Preservation

In high-growth e-commerce and retail environments, customer acquisition budgets are frequently squandered when acquired customers churn within 90 days. RetainIQ provides companies with a deterministic financial guardrail. By identifying the **339 accounts in the crisis tier ($\geq 80\%$)**, customer success representatives can proactively apply targeted retention incentives (such as personalized loyalty credits or service check-ins) to preserve recurring revenue before cancellation occurs. This reduces wasteful blanket marketing discounts, protecting operating margins.

5.2 Operational Productivity for Support Teams

In traditional organizations, support teams spend up to 35% of their weekly hours manually querying databases, merging spreadsheets, and determining which accounts require attention. RetainIQ automates this entire discovery cycle: team members arrive in the morning with a pre-prioritized, pre-filtered action queue waiting in their inboxes, allowing staff to spend 100% of their time on high-touch customer problem resolution rather than manual data entry.

5.3 Empowering MSMEs & Democratizing Enterprise Analytics

Sophisticated enterprise customer retention platforms (like Salesforce Einstein or Adobe Experience Platform) cost tens of thousands of dollars in licensing fees, placing them out of reach for small-and-medium enterprises (MSMEs) and regional startups. RetainIQ was constructed entirely using open-source, production-grade technologies (Python, SQLAlchemy, Scikit-Learn, Airflow, and Streamlit). It demonstrates how regional enterprises can deploy Fortune-500 level predictive retention intelligence with zero software licensing overhead.



6. Learning and Reflection

6.1 Technical Learnings & Engineering Growth

- **Relational Database Design & SQL Optimization:** Transitioned from writing simple queries to designing normalized database schemas, building complex SQL analytical views, and enforcing data integrity across thousands of transactional records using SQLAlchemy.
- **Statistical Feature Engineering:** Learned that feeding raw columns directly into machine learning models yields poor results. Mastered behavioral RFM quintile analysis (pd.qcut), feature imputation, and one-hot encoding to capture customer velocity and engagement.
- **Production Machine Learning & Model Calibration:** Learned that classification accuracy is a misleading metric in imbalanced business datasets. Mastered threshold tuning, focusing on Recall, Precision-Recall tradeoffs, and continuous probability scoring via predict_proba.
- **Workflow Automation with Apache Airflow:** Gained hands-on expertise with modern Airflow 2.x TaskFlow decorators (@dag, @task), task dependency chains, retry policies with exponential backoff, and avoiding XCom data bloat through database pushdown patterns.
- **Enterprise API Integration:** Learned production authentication protocols using Google OAuth 2.0, token management (credentials.json and token.json), and programmatic MIME multipart HTML email dispatch.
- **Decoupled Business Intelligence:** Understood the architectural distinction between transactional databases (OLTP) and analytical reporting marts (OLAP), designing decoupled CSV data contracts for Microsoft Power BI.

6.2 Management, Problem-Solving & Project Experience

- **Before the Internship:** My knowledge was largely academic and compartmentalized. I knew how to write isolated Python functions, run queries in SQLite, and build simple visuals in Power BI, but I lacked an understanding of how real-world data systems integrate end-to-end.
- **After the Internship:** Through the guidance of my mentor **Nithyasri kannathal EL** and the structured curriculum of **SURE Trust ProEd**, I experienced the complete software development lifecycle (SDLC). I learned how to decompose an ambiguous business problem ("*How do we stop customers from leaving?*"), architect an automated multi-tier solution, conduct rigorous automated testing (Pytest), and present technical deliverables to executive leadership.



7. Conclusion and Future Scope

7.1 Recap of Objectives & Achievements

The RetainIQ project successfully achieved all core research and engineering milestones:

1. **Automated Pipeline:** Engineered a 5-stage automated MLOps pipeline covering SQL ingestion, RFM feature engineering, Random Forest inference, priority alert emailing, and Power BI CSV synchronization.
2. **Predictive Accuracy:** Achieved **81.8% accuracy** and an **0.86 ROC-AUC**, isolating 339 high-risk accounts exceeding the 80% decision threshold.
3. **Decoupled BI Integration:** Built an interactive Power BI Control Tower and a Streamlit Operations Console that refresh analytical views in under 1 second without database locks.
4. **Autonomous Execution:** Integrated Apache Airflow for hands-free daily execution at 6:00 AM, alongside real-world Gmail OAuth alert dispatch.

7.2 Future Scope & Enhancements

1. **Cloud Warehouse Migration:** Scale the storage and compute layer from SQLite to cloud-native data warehouses (**Snowflake, Google BigQuery, or Amazon Redshift**) using partitioned columnar tables to support millions of customer accounts.
2. **Multi-Channel Retention Dispatch:** Expand the alerting subsystem beyond Gmail OAuth to incorporate direct **WhatsApp Business API** notifications and automated **Twilio SMS** loyalty discount dispatch.
3. **Automated A/B Uplift Modeling:** Transition from passive churn prediction to **Causal Uplift Modeling** (predicting which customers will change their behavior *specifically because* they received a retention offer versus those who would stay anyway).
4. **Natural Language Sentiment Analysis:** Ingest unstructured customer support transcripts and chat logs using Large Language Models (LLMs) to detect negative sentiment, complaints, and product frustration before numerical metrics reflect the issue.